

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418066
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Park; Yoon Woo

Battery module having improved energy density and battery pack including the same

Abstract

A battery module includes a unit module stack, in which unit modules, each of which includes one or more battery cells and a cell cartridge configured to receive the battery cells therein, are horizontally or vertically stacked on the basis of the ground; a pair of insulation sheets located at outermost sides of the unit module stack; and a pair of end plates located outside the pair of insulation sheets, wherein each of the pair of end plates is provided with a first protuberance protruding in a direction in which the unit module stack is located, and each of the pair of insulation sheets is provided with a first receiving portion configured to receive the first protuberance of a corresponding one of the end plates, the first receiving portion having an identical shape to the first protuberance, and a battery pack including the same.

Inventors:	Park; Yoon Woo (Daejeon, KR)
Applicant:	LG ENERGY SOLUTION, LTD. (Seoul, KR)
Family ID:	1000008821801
Assignee:	LG ENERGY SOLUTION, LTD. (Seoul, KR)
Appl. No.:	17/788412
Filed (or PCT Filed):	November 26, 2020
PCT No.:	PCT/KR2020/016905
PCT Pub. No.:	WO2021/182716
PCT Pub. Date:	September 16, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230035203 A1	Feb. 02, 2023

Foreign Application Priority Data

KR

10-2020-0030584

Mar. 12, 2020

Publication Classification

Int. Cl.: H01M50/204 (20210101); H01M50/505 (20210101); H01M50/557 (20210101)

U.S. Cl.:

CPC H01M50/204 (20210101); H01M50/505 (20210101); H01M50/557 (20210101);

Field of Classification Search

CPC: H01M (2220/20); H01M (50/204); H01M (50/209); H01M (50/249); H01M (50/289); H01M (50/291); H01M (50/505); H01M (50/557); Y02E (60/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0325059	12/2008	Niedzwiecki et al.	N/A	N/A
2012/0052359	12/2011	Yoshitake et al.	N/A	N/A
2015/0064546	12/2014	Murakami et al.	N/A	N/A
2016/0141712	12/2015	Choi et al.	N/A	N/A
2017/0141365	12/2016	Kim et al.	N/A	N/A
2017/0194681	12/2016	Kim et al.	N/A	N/A
2017/0309869	12/2016	Kim et al.	N/A	N/A
2017/0358784	12/2016	Jeong et al.	N/A	N/A
2018/0040863	12/2017	Kim et al.	N/A	N/A
2018/0159096	12/2017	Kim et al.	N/A	N/A
2021/0036270	12/2020	Motohashi et al.	N/A	N/A
2021/0305646	12/2020	Kubota et al.	N/A	N/A
2023/0098629	12/2022	Fukuda et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107710450	12/2017	CN	N/A
211062766	12/2019	CN	N/A
211320161	12/2019	CN	N/A
3 168 899	12/2016	EP	N/A
2011-528729	12/2010	JP	N/A
2014-10983	12/2013	JP	N/A
2015-50067	12/2014	JP	N/A
2015-111493	12/2014	JP	N/A
2016-46207	12/2015	JP	N/A
2018-46233	12/2015	JP	N/A
2018-54130	12/2017	JP	N/A
2018-56087	12/2017	JP	N/A

2019-186040	12/2018	JP	N/A
2020-107462	12/2019	JP	N/A
10-2012-0022922	12/2011	KR	N/A
10-2014-0147555	12/2013	KR	N/A
10-2015-0115251	12/2014	KR	N/A
10-2015-0137262	12/2014	KR	N/A
10-2017-0065864	12/2016	KR	N/A
10-2017-0138746	12/2016	KR	N/A
WO 2019/198518	12/2018	WO	N/A
WO 2021/199493	12/2020	WO	N/A

OTHER PUBLICATIONS

Machine translation KR20140147555A (Year: 2014). cited by examiner

Extended European Search Report for European Application No. 20923951.6, dated Dec. 20, 2023. cited by applicant

International Search Report (PCT/ISA/210) issued in PCT/KR2020/016905 mailed on Mar. 3, 2021. cited by applicant

Primary Examiner: Lynch; Victoria H

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

TECHNICAL FIELD

(1) This application claims the benefit of priority to Korean Patent Application No. 2020-0030584 filed on Mar. 12, 2020, the disclosure of which is incorporated herein by reference in its entirety.

(2) The present invention relates to a battery module having improved energy density and a battery pack including the same, and more particularly to a battery module configured such that a support portion is formed at an insulation sheet, whereby it is possible to further receive a battery cell without a cell cartridge and a battery pack including the same.

BACKGROUND ART

(3) Required capacities of secondary batteries used as energy sources of various kinds of electronic devices inevitably used in modern society have been increased due to an increase in usage of mobile devices, increasing complexity of the mobile devices, and development of electric vehicles. In order to satisfy demand of users, a plurality of battery cells is disposed in a small-sized device, whereas a battery module including a plurality of battery cells electrically connected to each other is used in a vehicle.

(4) In the battery module, a plurality of secondary batteries, i.e. battery cells, is connected to each other in series or in parallel in order to increase capacity and output of the battery module, and various research has been conducted to reduce a dead space in the battery module or to increase the capacity of each battery cell in order to increase the capacity of the battery module.

(5) FIG. 1 is an exploded perspective view of conventional battery module. As shown in FIG. 1, a pair of insulation sheets **20** are located inside a pair of end plates **10** located at the outermost sides of the conventional battery module, and a plurality of unit modules, each of which includes at least one battery cell and a cell cartridge configured to receive the battery cell, is stacked between the pair of insulation sheets **20** to constitute a unit module stack **30**.

(6) In the case in which a battery cell is added in order to increase the capacity of the battery

module, therefore, a cell cartridge is necessarily needed, which increases the weight of the battery module or a battery pack, whereby energy density thereof is reduced.

(7) Also, in the conventional battery module, each of the end plates **10** and a corresponding one of the insulation sheets **20** are disposed simply in surface contact with each other without being fixed, whereby it is not easy to assemble the battery module.

(8) In Japanese Patent Application Publication No. 2016-046207, an insulation plate is provided between an end plate and a battery holder, and the insulation plate is fixed by connection therebetween through an insertion portion. However, the above problem has still not been solved.

PRIOR ART DOCUMENT

(9) (Patent Document 1) Japanese Patent Application Publication No. 2016-046207

DISCLOSURE

Technical Problem

(10) The present invention has been made in view of the above problems, and it is an object of the present invention to provide a battery module configured such that stacking of battery cells is possible without addition of a cell cartridge and a battery pack including the same.

(11) It is another object of the present invention to provide a battery module configured such that an end plate and an insulation sheet can be fixed to each other and a battery pack including the same.

Technical Solution

(12) In order to accomplish the above objects, a battery module according to the present invention includes a unit module stack, in which unit modules include one or more battery cells and a cell cartridge configured to receive the one or more battery cells therein, are horizontally or vertically stacked in a first direction; a pair of insulation sheets located at outermost sides of the unit module stack; and a pair of end plates located outside the pair of insulation sheets, wherein each of the pair of end plates is provided with a first protuberance protruding in the first direction, and each of the pair of insulation sheets is provided with a first receiving portion configured to receive the first protuberance of a corresponding one of the end plates, the first receiving portion having a complimentary shape to the first protuberance.

(13) Also, in the battery module according to the present invention, the first protuberance may be a pair of first protuberances provided in a vicinity of each side of an upper edge of the end plate and the first receiving portion may be pair of first receiving portions.

(14) Also, in the battery module according to the present invention, each of the pair of end plates may be provided with a second protuberance protruding in the first direction, each of the pair of insulation sheets may be provided with a second receiving portion configured to receive the second protuberance of a corresponding one of the end plates, the second receiving portion having a complimentary shape to the second protuberance, the second protuberance may be located between the pair of first protuberances, and the second receiving portion may be located between the pair of first receiving portions.

(15) Also, in the battery module according to the present invention, each of the second protuberance and the second receiving portion may be configured to have the shape of a bar extending by a predetermined length in the first direction.

(16) Also, in the battery module according to the present invention, each of the pair of end plates may be provided with a third protuberance protruding in the first direction, and each of the pair of insulation sheets may be provided with a third receiving portion configured to receive the third protuberance of a corresponding one of the end plates, the third receiving portion having a complimentary shape to the third protuberance.

(17) Also, in the battery module according to the present invention, the third protuberance may be formed along the side edge of the end plate.

(18) Also, in the battery module according to the present invention, the third protuberance may be formed along the side edge and the lower edge of the end plate **400** so as to have an L-shape.

(19) Also, in the battery module according to the present invention, a double-sided tape having a predetermined size may be provided above the second protuberance of the end plate so as to be brought into contact with the insulation sheet.

(20) Also, in the battery module according to the present invention, the first receiving portion, the second receiving portion, and the third receiving portion of the insulation sheet may protrude in the first direction.

(21) Also, in the battery module according to the present invention, one of the battery cells in direct contact with the insulation sheet may be supported by the first receiving portion, the second receiving portion, and the third receiving portion of the insulation sheet.

(22) Also, in the battery module according to the present invention, the end plate may be provided at opposite sides of the upper edge thereof with a pair of first fastening members, each of the first fastening members having a first fastening hole, through which a fastening rod extends, formed therein, and the first fastening member may be formed so as to extend above the upper edge of the insulation sheet.

(23) Also, in the battery module according to the present invention, each cell cartridge of the unit modules may be provided at opposite sides of the upper edge thereof with a pair of second fastening members, each of the second fastening members having a second fastening hole, through which the fastening rod extends, formed therein.

(24) In addition, a battery pack according to the present invention includes the battery module.

(25) An outer surface of a side of each cell cartridge of the unit modules may be provided with a first projection protruding by a predetermined length and a first recess.

(26) In the present invention, one or more constructions that do not conflict with each other may be selected and combined from among the above constructions.

Advantageous Effects

(27) A battery module according to the present invention and a battery pack including the same have a merit in that an end plate and an insulation sheet are provided with a plurality of protuberances and a plurality of receiving portions, respectively, whereby fixing therebetween is possible and assembly is more easily achieved.

(28) In addition, the battery module according to the present invention and the battery pack including the same have an advantage in that the plurality of receiving portions is formed in the insulation sheet, whereby it is possible to further stack battery cells without a separate cell cartridge, and therefore it is possible to improve energy density of the battery module.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an exploded perspective view of a conventional battery module.

(2) FIG. 2 is a perspective view of a battery pack according to the present invention.

(3) FIG. 3 is an exploded perspective view of the battery pack according to the present invention.

(4) (a) of FIG. 4 is a plan view of a lower housing according to the present invention and a battery module received therein, and (b) of FIG. 4 is a bottom view of an upper housing.

(5) FIG. 5 is a perspective view of a battery module according to the present invention.

(6) FIG. 6 is an exploded perspective view of the battery module according to the present invention.

(7) FIG. 7 is a perspective view of cell cartridges according to the present invention when viewed from one side.

(8) FIG. 8 is a perspective view of the cell cartridges according to the present invention when viewed from the other side.

(9) FIG. 9 is a perspective view of the cell cartridges according to the present invention when

viewed from above.

(10) FIG. **10** is an enlarged view showing the upper parts of the cell cartridges according to the present invention.

(11) FIG. **11** is a perspective view of an insulation sheet and an end plate according to the present invention.

BEST MODE

(12) In the present application, it should be understood that the terms “comprises,” “has,” “includes,” etc. specify the presence of stated features, numbers, steps, operations, elements, components, or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, steps, operations, elements, components, or combinations thereof.

(13) In addition, the same reference numbers will be used throughout the drawings to refer to parts that perform similar functions or operations. In the case in which one part is said to be connected to another part in the specification, not only may the one part be directly connected to the other part, but also, the one part may be indirectly connected to the other part via a further part. In addition, that a certain element is included does not mean that other elements are excluded, but means that such elements may be further included unless mentioned otherwise.

(14) Hereinafter, a battery module having improved energy density according to the present invention and a battery pack including the same will be described with reference to the accompanying drawings.

(15) FIG. **2** is a perspective view of a battery pack according to the present invention, FIG. **3** is an exploded perspective view of a preferred battery pack according to the present invention, (a) of FIG. **4** is a plan view of a lower housing according to the present invention and a battery module received therein, and (b) of FIG. **4** is a bottom view of an upper housing.

(16) Referring to FIGS. **2** to **4**, the battery pack according to the present invention includes a lower housing **100**, an upper housing **200** coupled to the lower housing **100**, and an output terminal **300** protruding outwards through the upper housing **200** in a state of being connected to the lower housing **100** so as to electrically connect the battery pack to an external device.

(17) The lower housing **100** may include one or more battery modules, an ICB assembly, and a BMS assembly. The battery module includes a plurality of battery cells and cell cartridges. The battery module will be described below.

(18) Meanwhile, the ICB assembly serves to electrically connect the one or more battery modules to each other via an ICB busbar provided therein. The BMS assembly is coupled to the upper end of the ICB assembly in order to protect the battery pack. The ICB assembly and the BMS assembly are commonly known constructions, and therefore a more detailed description thereof will be omitted.

(19) As can be seen from FIGS. **3** and **4**, the upper housing **200** is coupled to the upper part of the lower housing **100**, in which the one or more battery modules, the ICB assembly, and the BMS assembly are received. The lower surface of the upper housing **200**, i.e. the inner bottom surface of the upper housing **200**, is provided with a plurality of fastening protrusions **210**, which is inserted into third fastening holes **634** of the cell cartridges **600** in order to fix the cell cartridges **600**.

(20) Specifically, the fastening protrusions **210** and the third fastening holes **634** of the cell cartridges **600** are provided at corresponding positions such that the fastening protrusions are located respectively in the third fastening holes when the upper housing **200** is coupled to the lower housing **100**. It is preferable that the fastening protrusions be formed in the same numbers as the third fastening holes **634**.

(21) The fastening protrusions **210** provided on the upper housing **200** may be configured to have the same shape or to have different shapes. In the case in which the fastening protrusions **210** have the same shape, it is possible to easily manufacture the fastening protrusions. In the case in which the fastening protrusions **210** have different shapes, it is possible to adjust coupling force of each fastening protrusion **210** based on the position thereof.

(22) Meanwhile, although the fastening protrusions **210** and the third fastening holes **634** may have a protrusion shape and a hole shape, respectively, the fastening protrusions **210** may have a hole shape and the third fastening holes **634** may have a protrusion shape, irrespective of the names thereof.

(23) A battery module according to the present invention received in the lower housing **100** will be described with reference to FIGS. **5** and **6**.

(24) FIG. **5** is a perspective view of a battery module according to the present invention, and FIG. **6** is an exploded perspective view of the battery module according to the present invention.

(25) The battery module according to the present invention may include a unit module stack, in which unit modules, each of which includes one or more battery cells **700** and a cell cartridge **600** configured to receive the battery cells **700** therein, are horizontally or vertically stacked on the basis of the ground, a pair of insulation sheets **500** located at the outermost sides of the unit module stack, and a pair of end plates **400** located outside the pair of insulation sheets **500**.

(26) Specifically, an end plate **400**, an insulation sheet **500**, a battery cell **700**, a battery cell **700**, a cell cartridge **600**, a battery cell **700**, a battery cell **700**, a cell cartridge **600**, a battery cell **700**, a battery cell **700**, an insulation sheet **500**, and an end plate **400** may be stacked in the battery module in that order.

(27) Meanwhile, each battery cell **700** may include an electrode assembly, an electrode lead protruding from the electrode assembly, and a case.

(28) The electrode assembly may be a jelly-roll type electrode assembly, which is configured to have a structure in which a long sheet type positive electrode and a long sheet type negative electrode are wound in the state in which a separator is interposed between the positive electrode and the negative electrode, a stacked type electrode assembly including unit cells, each of which is configured to have a structure in which a rectangular positive electrode and a rectangular negative electrode are stacked in the state in which a separator is interposed therebetween, a stacked and folded type electrode assembly, which is configured to have a structure in which unit cells are wound using a long separation film, or a laminated and stacked type electrode assembly, which is configured to have a structure in which unit cells are stacked in the state in which a separator is interposed therebetween and are then attached to each other. However, the present invention is not limited thereto.

(29) The electrode assembly is mounted in the case. The case is generally configured to have a laminate sheet structure including an inner layer, a metal layer, and an outer layer. The inner layer is disposed in direct contact with the electrode assembly, and therefore the inner layer must exhibit high insulation properties and high resistance to an electrolytic solution. In addition, the inner layer must exhibit high sealability in order to hermetically seal the case from the outside, i.e. a thermally-bonded sealed portion between inner layers must exhibit excellent thermal bonding strength. The inner layer may be made of a material selected from among a polyolefin-based resin, such as polypropylene, polyethylene, polyethylene acrylate, or polybutylene, a polyurethane resin, and a polyimide resin, which exhibit excellent chemical resistance and high sealability. However, the present invention is not limited thereto, and polypropylene, which exhibits excellent mechanical-physical properties, such as tensile strength, rigidity, surface hardness, and resistance to impact strength, and excellent chemical resistance, is the most preferably used.

(30) The metal layer, which is disposed so as to abut the inner layer, corresponds to a barrier layer configured to prevent moisture or various kinds of gas from permeating into the battery from the outside. Aluminum foil, which is light and easily shapeable, may be used as a preferred material for the metal layer.

(31) The outer layer is provided on the other surface of the metal layer. The outer layer may be made of a heat-resistant polymer that exhibits excellent tensile strength, resistance to moisture permeation, and resistance to air transmission such that the outer layer exhibits high heat resistance and chemical resistance while protecting the electrode assembly. As an example, the outer layer

may be made of nylon or polyethylene terephthalate. However, the present invention is not limited thereto.

(32) The electrode lead may be configured to have a structure in which the electrode lead is exposed outwards from the case in the state in which electrode tabs of the electrode assembly are electrically connected to the electrode lead or may be configured to have a structure in which the electrode lead directly connects the electrode assembly to the outside of the case in the state in which there are no electrode tabs. However, the present invention is not limited thereto. The battery cells correspond to commonly known constructions, and therefore a more detailed description thereof will be omitted.

(33) A plurality of battery cells **700** may be stacked through the cell cartridges **600**. The cell cartridges **600** are shown in FIGS. **7** to **10**.

(34) FIG. **7** is a perspective view of cell cartridges according to the present invention when viewed from one side, and FIG. **8** is a perspective view of the cell cartridges according to the present invention when viewed from the other side.

(35) In the battery module according to the present invention, the cell cartridges **600** may have the same external shape and size. That is, each of the cell cartridges **600** according to the present invention may include a pair of vertical frames **610** opposite each other in a state of being spaced apart from each other by a predetermined distance, a pair of horizontal frames **620** configured respectively to connect upper ends and lower ends of the pair of vertical frames **610** to each other, and a pair of second fastening members **630** formed in the vicinity of corners at which the upper ends of the pair of vertical frames **610** and a corresponding one of the horizontal frames **620** join so as to extend upwards by a predetermined height.

(36) First, the vertical frames **610** will be described in more detail. Each vertical frame **610**, which has an approximately flat structure, is provided with a first coupling member **611** and a second coupling member **612** formed so as to face in one direction in a state of being spaced apart from each other by a predetermined distance. In the pair of vertical frames **610**, the first coupling members **611** and the second coupling members **612** may be located in diagonal directions.

(37) That is, as shown in FIGS. **7** and **8**, in the cell cartridges **600** according to the present invention, the first coupling members **611** and the second coupling members **612** are located in diagonal directions such that the same kinds of cell cartridges can be coupled to each other in a state of facing each other.

(38) The first coupling member **611** includes a first coupling protrusion **611(a)** protruding by a predetermined length more than the edge of the vertical frame **610** and a first coupling tight-contact recess **611(b)** formed in the outer surface of the first coupling protrusion **611(a)** so as to be depressed by a predetermined depth. The second coupling member **612** may include a second coupling protrusion **612(a)** seated in the first coupling tight-contact recess **611(b)** in a state of protruding by a predetermined length more than the edge of the vertical frame **610** and a second coupling insertion recess **612(b)** configured to receive the first coupling protrusion **611(a)**.

(39) Consequently, when a plurality of battery cells **700** is received in an inner space defined by the vertical and horizontal frames **610** and **620** of one cell cartridge **600** and then a pair of cell cartridges **600** is brought into tight contact with each other, the second coupling protrusions **612(a)** are seated in the first coupling tight-contact recesses **611(b)** and the first coupling protrusions **611(a)** are inserted into the second coupling insertion recesses **612(b)**, whereby a unit module stack capable of inhibiting movement of the cell cartridges **600** upwards, downwards, leftwards, and rightwards may be obtained.

(40) Meanwhile, each of the pair of vertical frames **610** and a lower one of the horizontal frames **620** are provided with a first inner extension portion **613** and a second inner extension portion **623**, respectively. When two battery cells **700** are received in a cell cartridge **600**, these extension portions partition the battery cells **700** from each other.

(41) FIG. **9** is a perspective view of the cell cartridges according to the present invention when

viewed from above. In order to increase the force of coupling between the cell cartridges **600**, the horizontal frame **620** may be further provided with a third coupling member **621** and a fourth coupling member **622**. Specifically, the third coupling member **621** and the fourth coupling member **622** are located at the lower horizontal frame **620**. The third coupling member **621** and the fourth coupling member **622** protrude in one direction, more specifically in the same direction as the direction in which the first coupling member **611** and the second coupling member **612** protrude, and are located in a state of being spaced apart from each other by a predetermined distance.

(42) Here, the third coupling member **621** includes a third coupling protrusion protruding by a predetermined length more than the edge of the horizontal frame **620** and a third coupling tight-contact recess formed in the outer surface of the third coupling protrusion so as to be depressed by a predetermined depth. The fourth coupling member **622** may include a fourth coupling protrusion seated in the third coupling tight-contact recess in a state of protruding by a predetermined length more than the edge of the horizontal frame **620** and a fourth coupling insertion recess configured to receive the third coupling protrusion. These coupling members are identical in external appearance and coupling structure to the first coupling members **611** and the second coupling members **612**, and therefore a further description thereof will be omitted.

(43) FIG. **10** is an enlarged view showing the upper parts of the cell cartridges according to the present invention. In each of the cell cartridges **600** according to the present invention, the pair of second fastening members **630** is located in the vicinity of corners at which the upper ends of the pair of vertical frames **610** and the upper horizontal frame **620** join, and protrudes upwards more than the upper horizontal frame **620**.

(44) The pair of second fastening members **630** is configured to fix the cell cartridges **600**, the battery cells **700**, the insulation sheets **500**, and the end plates **400** to each other in the state in which the battery cells, the insulation sheets, and the end plates are stacked in front of and at the rear of the cell cartridges.

(45) Specifically, each of the second fastening members **630** may include a second fastening hole **631**, a fifth coupling protrusion **632**, a step portion **633**, and a third fastening hole **634**.

(46) The second fastening hole **631** provides a space through which a fastening rod **800** extends, and is formed in a horizontal direction corresponding to the front and the rear of the cell cartridge **600**.

(47) The fifth coupling protrusion **632** and the step portion **633** are configured to increase the force of coupling between the pair of cell cartridges **600** when the cell cartridges are brought into tight contact with each other. The fifth coupling protrusion **632** extends outwards by a predetermined length along the outer diameter of the second fastening hole **631**, whereas the step portion **633** is formed outside the second fastening hole **631** opposite the side at which the fifth coupling protrusion **632** is formed. Of course, the inner diameter of the step portion **633** is greater than the inner diameter of the second fastening hole **631** so as not to interfere with the fastening rod **800**.

(48) Meanwhile, in the pair of second fastening members **630**, the fifth coupling protrusion **632** and the step portion **633** are located in opposite directions. That is, in the case in which one of the second fastening members **630** is formed such that the fifth coupling protrusion **632** faces the front and the step portion **633** faces the rear, the fifth coupling protrusion **632** of the other second fastening member **630** faces the rear and the step portion **633** of the other second fastening member faces the front.

(49) When the pair of cell cartridges **600** are brought into tight contact with each other, therefore, the fifth coupling protrusion **632** formed on any one of the cell cartridges **600** is inserted into the second fastening hole **631** of the other cell cartridge **600**, which is coupled to the one of the cell cartridges while facing each other.

(50) As described above, the third fastening hole **634** is configured to receive a corresponding one of the fastening protrusions **210** of the upper housing **200** (see FIGS. **4** and **9**), and is formed in the

upper end of the second fastening member **630**, i.e. in the second fastening member **630** in a vertical direction.

(51) Next, the end plate and the insulation sheet will be described. FIG. **11** is a perspective view of an insulation sheet and an end plate according to the present invention.

(52) First, the end plate **400** will be described. The end plate, which is located at each of the outermost sides of the battery module and which has an approximately flat structure, includes a first protuberance **410**, a second protuberance **420**, a third protuberance **430**, a double-sided tape **440**, a horizontal bent portion **450**, a vertical bent portion **460**, and a first fastening member **470**.

(53) The first protuberance **410** is provided in the vicinity of each side of the upper edge of the end plate **400**, and may minimize movement of the insulation sheet **500** when brought into tight contact therewith.

(54) That is, the first protuberance **410** is configured to have a projection formed at the lower part thereof while protruding in a direction toward the insulation sheet **500**.

(55) The second protuberance **420**, which is configured to have the shape of a long bar having a predetermined length in the horizontal direction, is located between the pair of first protuberances **410**, and is formed so as to protrude in the direction toward the insulation sheet **500**, in the same manner as the first protuberance **410**.

(56) A pair of third protuberances **430** may be formed along the side edge of the end plate **400**. For example, each of the third protuberances may be formed in the vicinity of a corner at which the vertical edge and the horizontal edge of the end plate **400** join so as to have an L-shape.

(57) The double-sided tape **440** is located in the vicinity of the second protuberance **420**, e.g. above the second protuberance **420**, to increase the force of tight contact with the insulation sheet **500**.

(58) A plurality of horizontal bent portions **450** and a plurality of vertical bent portions **460** are configured to improve heat dissipation through an increase in surface area and at the same time to prevent deformation of the battery module when impact is applied thereto leftwards, rightwards, upwards, and downwards.

(59) The first fastening member **470** is provided at opposite sides of the upper edge of the end plate, and has a first fastening hole **471**, through which the fastening rod **800** extends, formed therein. Here, it is preferable that the first fastening member **470** be formed so as to extend upwards more than the upper edge of the insulation sheet **500**.

(60) The insulation sheet **500** is located at the inner surface of each of the pair of end plates **400** and has an approximately flat structure. One surface of the insulation sheet is brought into tight contact with the end plate **400**, and the other surface of the insulation sheet abuts the battery cell **700**.

(61) Specifically, the insulation sheet **500** includes first receiving portions **510**, a second receiving portion **520**, and third receiving portions **530**. The first receiving portions **510** receive the first protuberances **410** of the end plate **400**, the second receiving portion **520** receives the second protuberance **420** of the end plate **400**, and the third receiving portions **530** receive the third protuberances **430** of the end plate **400**.

(62) When the end plate **400** and the insulation sheet **500** are brought into tight contact with each other, therefore, the side edges of the end plate and the insulation sheet as well as the upper edges of the end plate and the insulation sheet are securely brought into tight contact with each other, whereby the force of coupling therebetween may be increased.

(63) Here, the first receiving portions **510**, the second receiving portion **520**, and the third receiving portions **530** of the insulation sheet **500** are identical in shape to the first protuberances **410**, the second protuberance **420**, and the third protuberances **430** of the end plate **400**, respectively. These receiving portions correspond to spaces configured to receive the protuberances and thus are in protruding states, although the term “receiving portions” is used.

(64) Meanwhile, the battery cell **700** may be seated in a space defined by the first receiving portions **510**, the second receiving portion **520**, and the third receiving portions **530** of the

insulation sheet **500**.

(65) That is, three edges of the battery cell **700** may be supported by the pair of third receiving portions **530**, each of which is formed so as to have an L-shape, and the other edge of the battery cell may be supported by the pair of the first receiving portions **510** and the second receiving portion **520**.

(66) Conventionally, a cell cartridge is needed to mount the battery cell **700**. However, the battery module according to the present invention has an advantage in that it is possible to further mount the battery cell **700** through the receiving portions of the insulation sheet **500**, whereby it is possible to increase energy density of the battery module.

(67) Meanwhile, the fastening rods **800** extend through the first fastening holes **471** of the end plates **400** and the second fastening holes **631** of the cell cartridges **600** and are then fixed.

(68) The battery pack having the above construction may be applied to various kinds of devices, for example, a vehicle, such as an electric vehicle or a hybrid electric vehicle.

(69) Although the specific details of the present invention have been described in detail, those skilled in the art will appreciate that the detailed description thereof discloses only preferred embodiments of the present invention and thus does not limit the scope of the present invention. Accordingly, those skilled in the art will appreciate that various changes and modifications are possible, without departing from the category and technical idea of the present invention, and it will be obvious that such changes and modifications fall within the scope of the appended claims.

DESCRIPTION OF REFERENCE SYMBOLS

(70) **10**: End plate **20**: Insulation sheet **30**: Unit module stack **100**: Lower housing **200**: Upper housing **210**: Fastening protrusion **300**: Output terminal **400**: End plate **410**: First protuberance **420**: Second protuberance **430**: Third protuberance **440**: Double-sided tape **450**: Horizontal bent portion **460**: Vertical bent portion **470**: First fastening member **471**: First fastening hole **500**: Insulation sheet **510**: First receiving portion **520**: Second receiving portion **530**: Third receiving portion **600**: Cell cartridge **610**: Vertical frame **611**: First coupling member **611(a)**: First coupling protrusion **611(b)**: First coupling tight-contact recess **612**: Second coupling member **612(a)**: Second coupling protrusion **612(b)**: Second coupling insertion recess **613**: First inner extension portion **620**: Horizontal frame **621**: Third coupling member **621(a)**: Third coupling protrusion **621(b)**: Third coupling tight-contact recess **622**: Fourth coupling member **622(a)**: Fourth coupling protrusion **622(b)**: Fourth coupling insertion recess **623**: Second inner extension portion **630**: Second fastening member **631**: Second fastening hole **632**: Fifth coupling protrusion **633**: Step portion **634**: Third fastening hole **700**: Battery cell **800**: Fastening rod

Claims

1. A battery module comprising: a unit module stack, in which unit modules comprising one or more battery cells and a cell cartridge configured to receive the one or more battery cells therein, are horizontally or vertically stacked in a first direction; a pair of insulation sheets located at outermost sides of the unit module stack; and a pair of end plates located outside the pair of insulation sheets, wherein each of the pair of end plates is provided with a first protuberance at an edge of each of the end plates protruding in the first direction, and wherein each of the pair of insulation sheets is provided with a first receiving portion at an edge of each of the pair of insulation sheets configured to receive the first protuberance of a corresponding one of the end plates, the first receiving portion having a complimentary shape to the first protuberance.
2. The battery module according to claim 1, wherein the first protuberance is a pair of first protuberances provided in a vicinity of each side of an upper edge of each end plate and the first receiving portion is a pair of first receiving portions.
3. The battery module according to claim 2, wherein each of the pair of end plates is provided with a second protuberance protruding in the first direction, wherein each of the pair of insulation sheets

- is provided with a second receiving portion configured to receive the second protuberance of a corresponding one of the end plates, the second receiving portion having a complimentary shape to the second protuberance, and wherein the second protuberance is located between the pair of first protuberances and the second receiving portion is located between the pair of first receiving portions.
4. The battery module according to claim 3, wherein each of the second protuberance and the second receiving portion is configured to have a shape of a bar extending by a predetermined length in the first direction.
 5. The battery module according to claim 4, wherein each of the pair of end plates is provided with a third protuberance protruding in the first direction, and wherein each of the pair of insulation sheets is provided with a third receiving portion configured to receive the third protuberance of a corresponding one of the end plates, the third receiving portion having a complimentary shape to the third protuberance.
 6. The battery module according to claim 5, wherein the third protuberance is formed along a side edge of the end plate.
 7. The battery module according to claim 5, wherein the third protuberance is formed along a side edge and a lower edge of the end plate so as to have an L-shape.
 8. The battery module according to claim 4, wherein a double-sided tape having a predetermined size is provided above the second protuberance of the end plate so as to be brought into contact with each of the pair of insulation sheets.
 9. The battery module according to claim 7, wherein the first receiving portion, the second receiving portion, and the third receiving portion of each of the pair of insulation sheets protrude in the first direction.
 10. The battery module according to claim 9, wherein one of the battery cells in direct contact with the insulation sheet is supported by the first receiving portion, the second receiving portion, and the third receiving portion of the insulation sheet.
 11. The battery module according to claim 10, wherein each end plate is provided at opposite sides of the upper edge thereof with a pair of first fastening members, each of the first fastening members having a first fastening hole, through which a fastening rod extends, formed therein, and wherein each first fastening member is formed so as to extend above an upper edge of the insulation sheet.
 12. The battery module according to claim 11, wherein each cell cartridge of the unit modules is provided at opposite sides of an upper edge thereof with a pair of second fastening members, each of the second fastening members having a second fastening hole, through which the fastening rod extends, formed therein.
 13. A battery pack comprising the battery module according to claim 1.
 14. The battery module according to claim 1, wherein an outer surface of a side of each cell cartridge of the unit modules is provided with a first projection protruding by a predetermined length and a first recess.
 15. The battery module according to claim 1, wherein each end plate is provided at opposite sides of the upper edge thereof with a pair of first fastening holes, the pair of first fastening holes being formed in the first protuberance.
 16. The battery module according to claim 1, wherein the first receiving portion is formed at a top edge of each of the pair of insulation sheets.
 17. The battery module according to claim 2, wherein the pair of first receiving portions is formed at a top edge of each of the pair of insulation sheets, and wherein the second receiving portion is between and below the pair of first receiving portions.
-